EXECUTIVE SUMMARY

Business Problem

Placed aims to improve Placement Success Rate by identifying the factors that contribute most to successful student placements. While some students with strong academic records fail to secure jobs, others with moderate academic performance achieve successful placements. This analysis investigates the key drivers behind these outcomes.

What We Did

• Performed data quality assessment and preprocessing
• Analyzed candidate academic performance
• Evaluated technical skill metrics
• Studied practical exposure indicators such as internships and projects
• Compared placed and non-placed students
• Conducted candidate archetype analysis
• Developed an interactive Tableau dashboard for stakeholder decision-making

Key Conclusions

• Technical skills have a stronger impact on placement success than academics alone
• Practical exposure significantly improves employability outcomes
• Placement success is driven by a combination of technical, practical, and professional competencies
• Candidate archetypes provide a structured way to identify placement potential

BUSINESS CONTEXT & NORTH STAR METRIC

About Placed

Placed is a hiring platform that connects students with recruiters and aims to improve hiring efficiency through better candidate identification and preparation.

North Star Metric

Placement Success Rate (**50.02%**)= Number of Students Placed ÷ Total Students

Why This Metric Matters

• Measures overall effectiveness of candidate placement efforts
• Directly impacts recruiter satisfaction
• Improves hiring efficiency and conversion rates
• Supports long-term business growth

Analysis Framework

Placement Success Rate

↓ Academics

↓ Technical Skills

↓ Practical Exposure

↓ Professional Competencies

↓ Placement Outcome

Why We Move to the Next Step

Before identifying placement drivers, it is important to validate the quality and reliability of the available data.

<u>HYPOTHESES & ANALYTICAL APPROACH</u>

Why We Defined Hypotheses

Before beginning the analysis, a set of hypotheses was created to identify the factors most likely to influence placement success. These hypotheses guided the analytical process and were later validated through exploratory data analysis.

Hypothesis H1

Students with stronger technical skills are more likely to be placed.

Reasoning:
Recruiters increasingly prioritize practical technical capabilities over academic performance alone.

Hypothesis H2

Students with internships, projects, and certifications achieve higher placement success rates.

Reasoning:
Practical exposure demonstrates job readiness and real-world application of skills.

Hypothesis H3

Academic performance positively influences placement outcomes but is not sufficient on its own.

Reasoning:
Many students possess strong academic records but lack industry-relevant skills.

Hypothesis H4

Communication, problem-solving, teamwork, and leadership skills improve employability outcomes.

Reasoning:
Recruiters assess workplace readiness in addition to technical capability.

Hypothesis H5

Candidates performing well across multiple dimensions achieve the highest placement success rates.

Reasoning:
Employability is influenced by a combination of academic, technical, practical, and professional competencies.

Expected Outcome

The analysis aims to determine which hypotheses are supported by the data and identify the strongest drivers of placement success.

DATA QUALITY ASSESSMENT

What We Checked

- Missing values
- Duplicate records
- Data consistency
- Outlier behaviour across important variables

What We Found

- No significant missing values were observed
- No duplicate records were identified
- Data quality was sufficient for analysis
- Outliers were reviewed and handled appropriately where required

Conclusion

The dataset was found to be reliable and suitable for exploratory analysis and business decision-making.

Why We Move to the Next Step

Since the dataset is trustworthy, the next step is to understand the overall characteristics of the student population.

UNDERSTANDING THE STUDENT POPULATION

What We Checked

- Academic performance levels
- Technical skill distribution
- Internship participation
- Project experience
- Professional skill indicators

What We Found

• Most students fall within moderate academic ranges
• Technical skill levels vary significantly across candidates
• Practical exposure differs widely among students
• Student profiles are highly diverse

Conclusion

The candidate pool consists of students with different strengths and weaknesses, indicating that multiple factors may influence placement outcomes.

Why We Move to the Next Step

Understanding the population alone is insufficient. We now need to identify which factors are most closely associated with placement success.

<u>PLACEMENT DRIVER ANALYSIS</u>

What We Checked

The relationship between candidate attributes and placement outcomes.

What We Found

Strong Positive Relationships:

• Technical skills
• Problem-solving ability
• Communication skills
• Internships
• Projects
• Coding proficiency

Weak Relationships:

• Age
• Family income
• Attendance

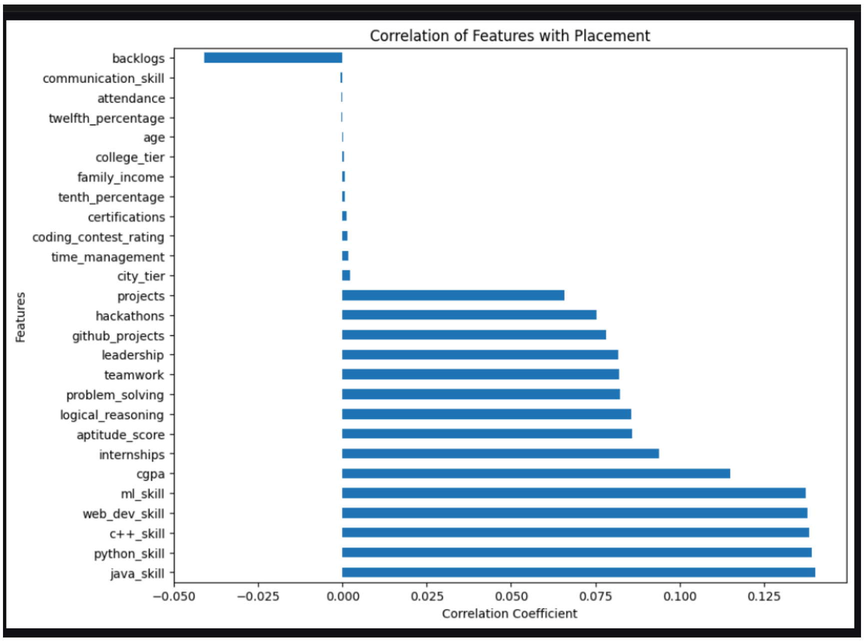
Key Insight

Placement success is more strongly associated with employability-related skills than demographic characteristics.

Why We Move to the Next Step

Correlation helps identify relationships, but it does not explain how placed and non-placed students differ in practice.

Screenshot Holder: Placement Correlation Heatmap



Description: Heatmap showing correlations between candidate attributes and placement status. This is one of the most important visuals in the report.

PLACED VS NON-PLACED STUDENT ANALYSIS

What We Checked

Comparison of average performance between placed and non-placed students.

What We Found

Placed students consistently demonstrate:

- Higher technical skill scores
- Better communication skills
- Greater practical exposure
- Stronger problem-solving abilities
- More project and internship experience

Conclusion

Successful candidates perform better across multiple employability dimensions rather than a single factor.

Why We Move to the Next Step

The next step is to understand whether academic performance alone can explain placement outcomes.

ACADEMIC PERFORMANCE VS TECHNICAL SKILLS

What We Checked

• Combined impact of academic performance and technical skills on placement outcomes

What We Found

• Strong technical skills improve placement outcomes even when academics are moderate

• High academic scores without strong technical skills produce weaker placement results

• Candidates possessing both strengths achieve the highest success rates

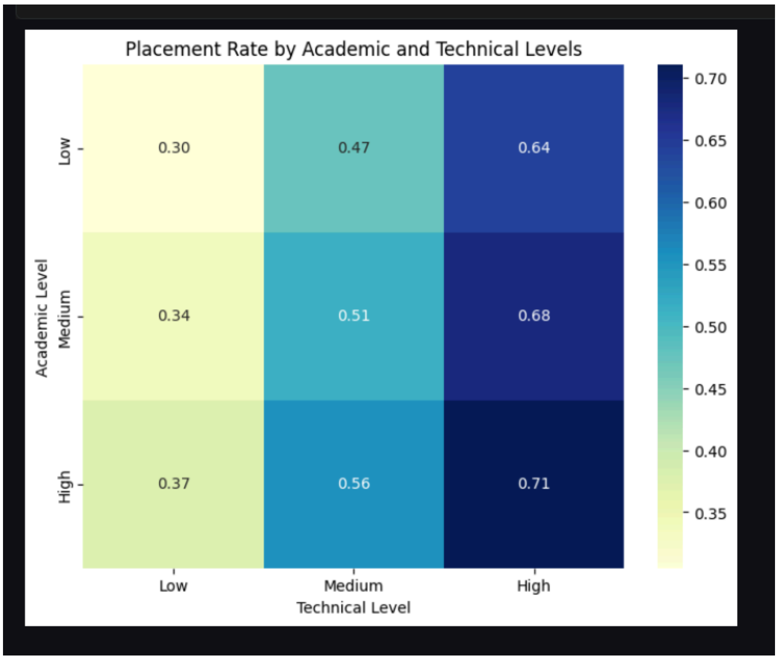
Key Insight

• Technical capability has a stronger impact on placement outcomes than academics alone

Why We Move Next

• Technical capability may become even more effective when supported by practical exposure

Screenshot Holder: Academic vs Technical Skills Heatmap



Description: Heatmap showing placement success across combinations of academic performance and technical skill levels.

<u>**TECHNICAL SKILLS VS PRACTICAL EXPOSURE**</u>

What We Checked

• Combined influence of technical skills and practical experience

What We Found

• Candidates with strong technical skills and practical exposure achieve the highest placement rates

• Projects and internships amplify the value of technical knowledge

• Practical exposure improves recruiter confidence

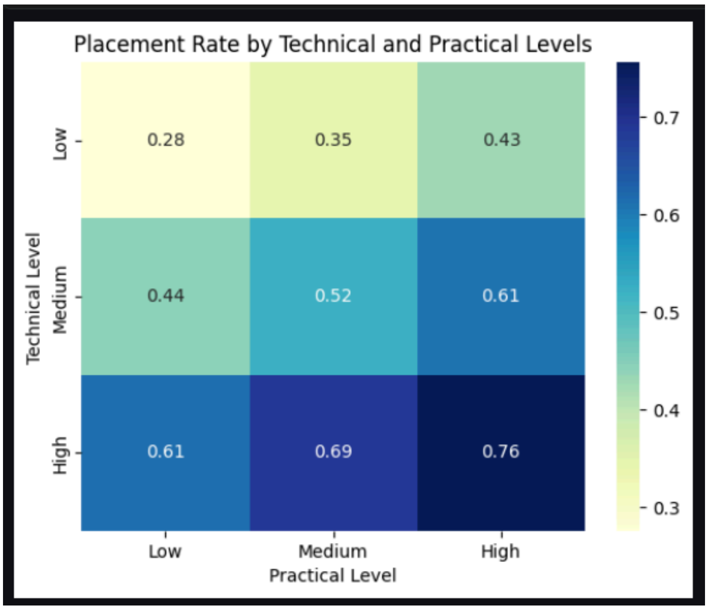
Key Insight

• Technical knowledge creates potential, while practical exposure demonstrates job readiness

Why We Move Next

• Placement outcomes are also influenced by professional competencies that recruiters value

Screenshot Holder: Technical Skills vs Practical Exposure Heatmap



Description: Heatmap illustrates placement success across different combinations of technical skills and practical experience.

Key Learning Before Segmentation

The previous analysis revealed that placement success is influenced by multiple dimensions rather than a single factor.

The strongest outcomes were observed among candidates who demonstrated:

• Strong technical capability

• Practical internship and project experience

• Effective communication and problem-solving skills

• Balanced overall employability profiles

CANDIDATE ARCHETYPE ANALYSIS

Why Archetypes Were Created

Recruiters evaluate candidates based on multiple characteristics simultaneously. Archetypes help identify recurring candidate profiles and placement trends.

Archetype Summary Table

(Insert the complete archetype table generated in the Python analysis.)

Key Observation

Different candidate groups demonstrate significantly different placement outcomes, highlighting the importance of a balanced skill profile.

ARCHETYPE INTERPRETATION & BUSINESS INSIGHTS

Elite Employability Profiles

Placement Rate: 82% – 87%

Characteristics

• Strong academics

• Strong technical skills

• High internship and project participation

• Strong communication and problem-solving abilities

Business Meaning

These candidates consistently demonstrate excellence across nearly all employability dimensions and achieve the highest placement success rates.

Recommendation

Recruiters should prioritize these candidates for high-value and competitive roles.

Technical-Core Achievers

Placement Rate: 70% – 83%

Characteristics

• Strong coding and technical capabilities

• Good project exposure

• Moderate academic performance

• High technical confidence

Business Meaning

These candidates demonstrate that strong technical skills can compensate for weaker academic performance.

Recommendation

Recruiters should avoid relying solely on academic filters and actively consider technically skilled candidates.

<u>Industry-Ready Practitioners</u>

Placement Rate: 55% – 77%

Characteristics

• Strong internships and project portfolios

• Good communication skills

• Practical problem-solving abilities

• Moderate technical foundation

Business Meaning

These candidates are well prepared for workplace environments due to their practical exposure and professional readiness.

Recommendation

Students in this group should strengthen technical skills to further improve placement outcomes.

<u>Academic-Oriented Profiles</u>

Placement Rate: 43% – 61%

Characteristics

• Strong academic performance

• Good educational foundation

• Limited practical exposure

• Weaker technical skill levels

Business Meaning

Academic excellence alone does not consistently translate into placement success.

Recommendation

Students should complement academic performance with projects, internships, and technical skill development.

High-Risk Employability Profiles

Placement Rate: 15% – 25%

Characteristics

• Weak technical skills

• Limited practical experience

• Lower professional readiness

• Multiple employability gaps

Business Meaning

These candidates face the greatest challenges in securing placements due to weaknesses across several critical dimensions.

Recommendation

Focused interventions, skill-building programs, and practical exposure opportunities are required to improve employability.

FINAL ARCHETYPE INSIGHT BOX

Screenshot Holder: Archetype Performance Comparison

Archetype	Placement Range	Core Characteristics
Elite Employability Profiles	*82% – 87%*	Strong across nearly all employability dimensions
Technical-Core Achievers	*70% – 83%*	Strong technical capability despite weaker academics
Industry-Ready Practitioners	*55% – 77%*	Strong practical exposure and professional readiness
Academic-Oriented	*43% – 61%*	Strong academics but

Profiles		weaker technical foundation
High-Risk Employability Profiles	*15% – 25%*	Weak technical and practical foundations across multiple dimensions

Description: Visual comparison of placement rates and characteristics across all identified archetypes.

Key Business Insight

The highest placement success rates are achieved by candidates who combine technical capability, practical exposure, and professional readiness. Academic excellence alone is not sufficient to maximize placement outcomes.

<u>TABLEAU DASHBOARD OVERVIEW</u>

Dashboard Objective

Provide recruiters and stakeholders with a centralized view of placement performance and candidate characteristics.

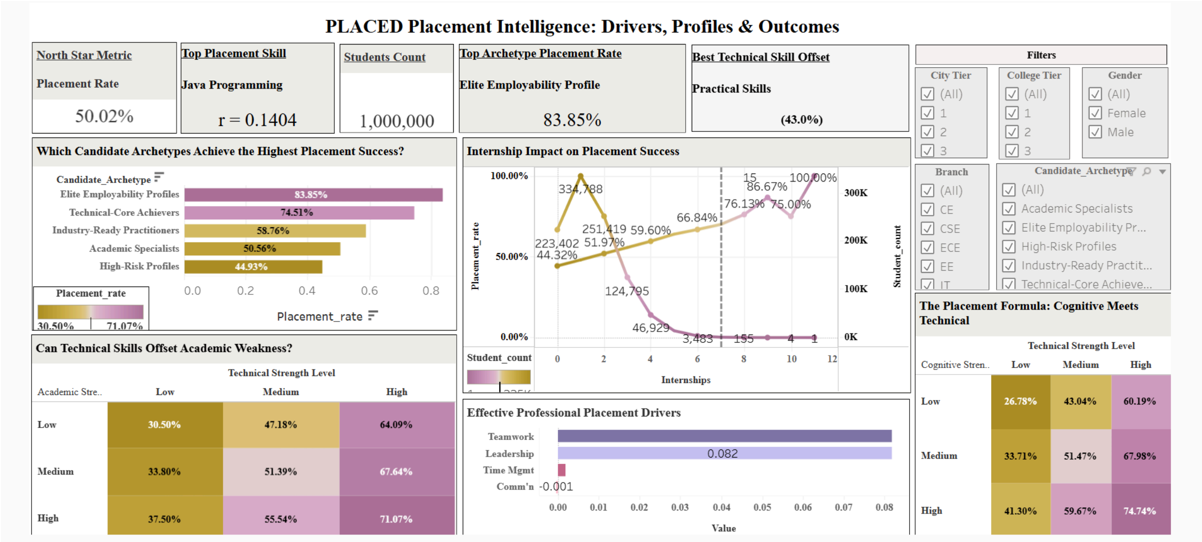
Dashboard Features

• Placement Success KPI
• Candidate Performance Monitoring
• Skill Analysis
• Placement Trends
• Candidate Segmentation

Business Value

The dashboard enables faster and more informed hiring decisions.

Screenshot Holder: Tableau Dashboard – Full Overview



Description: Full dashboard screenshot displaying KPIs, filters, placement metrics, and overall candidate analysis.

TABLEAU DASHBOARD ANALYSIS

Key Dashboard Components

• Placement Success Rate KPI
• Skill Performance Analysis
• Candidate Comparison Views
• Segment-Level Performance Insights

How Stakeholders Can Use It

• Identify high-potential candidates
• Monitor placement trends
• Evaluate candidate readiness
• Support recruiter decision-making

TABLEAU STORYBOARD & FINAL RECOMMENDATIONS

Storyboard Flow

[NL_PLACED | Tableau Public]

Business Problem

↓ Data Validation

↓ Placement Driver Analysis

↓ Academic Analysis

↓ Technical Skills Analysis

↓ Practical Exposure Analysis

↓ Candidate Archetypes

↓ Recommendations

Recommendations for Placed

• Develop an Employability Score that combines technical skills, practical exposure, and professional competencies.

• Prioritize skill-based shortlisting over academic-only filtering criteria.

• Use candidate archetypes to identify high-potential students and provide targeted interventions for at-risk profiles.

• Design personalized development programs for Academic-Oriented and High-Risk Employability Profiles to improve placement readiness.

Recommendations for Students

• Focus on building strong technical foundations through coding practice and domain-specific skills.

• Increase practical exposure by participating in internships, projects, hackathons, and certifications.

• Strengthen communication, teamwork, and problem-solving abilities to improve workplace readiness.

• Build a balanced employability profile rather than relying solely on academic performance.

Recommendations for Recruiters

• Reduce dependence on academic scores as the primary screening criterion.

• Adopt a multidimensional evaluation approach that includes technical, practical, and professional competencies.

• Consider Technical-Core Achievers and Industry-Ready Practitioners alongside academically strong candidates.

• Use employability-based segmentation to improve candidate selection efficiency and hiring outcomes.

Final Conclusion

Placement success is not determined by academic excellence alone. The analysis demonstrates that candidates who combine technical capability, practical exposure, professional readiness, and problem-solving skills consistently achieve the highest placement outcomes. By adopting a multidimensional approach to candidate evaluation and development, Placed can improve placement success rates, enhance recruiter satisfaction, and create better career outcomes for students.